

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

QUESTION 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Question

An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset:

118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318

Tasks

1. Construct an ECDF and a normal Q-Q plot.
2. Use both plots to evaluate the normality claim.
3. Identify evidence of skewness and/or extreme observations.
4. Recommend percentile-based limits or mean $\pm$ standard-deviation limits, with justification.

R Code

```
response_time <- c(
  118,121,125,127,129,131,134,136,138,141,
  144,147,151,156,162,171,184,205,249,318
)
```

```
# ECDF
```

```
plot(ecdf(response_time),
     main = "ECDF of Response Time",
     xlab = "Response Time (ms)",
     ylab = "Cumulative Probability")
```

```
# Normal Q-Q Plot
```

```
qqnorm(response_time,
       main = "Normal Q-Q Plot")
```

```
qqline(response_time, col = "red", lwd = 2)
```

```
# Statistics
```

```
mean(response_time)
```

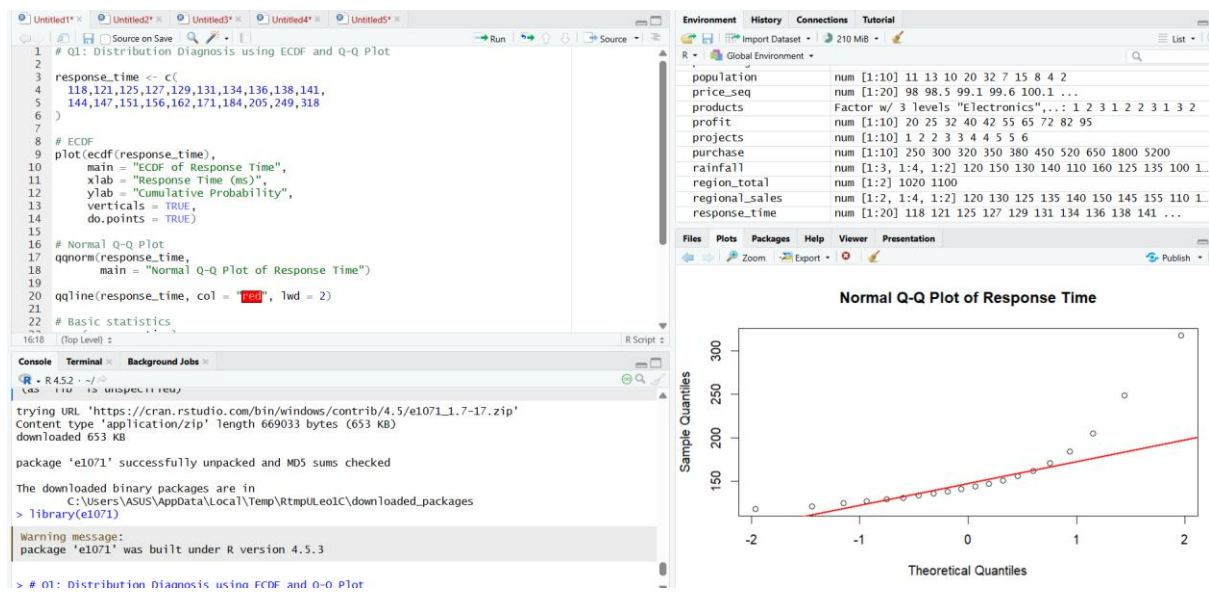
```
sd(response_time)
```

```
median(response_time)
```

```
# Percentile limits
```

```
quantile(response_time, c(0.05, 0.95))
```

OUTPUT



QUESTION 2 – Comparing Many Distributions and Bean Plots

Question

A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines.

Production Line Data:

- **L1:** 42, 44, 45, 46, 47, 48, 49, 50, 51, 53
- **L2:** 38, 41, 44, 47, 50, 53, 56, 59, 62, 65
- **L3:** 45, 46, 46, 47, 47, 48, 48, 49, 49, 50
- **L4:** 40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks

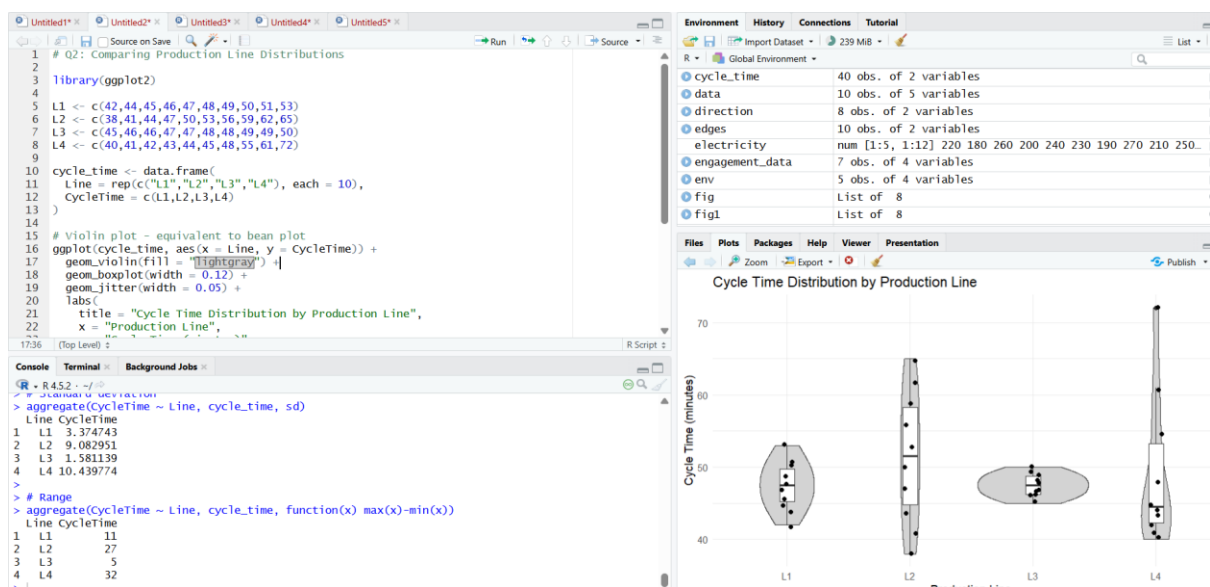
1. Create a visualization comparing all four distributions.
2. Construct bean plots or an equivalent distribution visualization along the vertical axis.
3. Determine which line has greatest variability and which has the most concentrated distribution.
4. Explain why comparing only means is insufficient.

R Code

```
library(ggplot2)
```

```
cycle_time <- data.frame(  
  Line = rep(c("L1", "L2",  
    "L3", "L4"), each = 10),  
  CycleTime = c(  
    42, 44, 45, 46, 47, 48, 49, 50, 51, 53,  
    38, 41, 44, 47, 50, 53, 56, 59, 62, 65,  
    45, 46, 46, 47, 47, 48, 48, 49, 49, 50,  
    40, 41, 42, 43, 44, 45, 48, 55, 61, 72  
  )  
)
```

```
ggplot(cycle_time, aes(x = Line, y = CycleTime)) +
  geom_violin() +
  geom_boxplot(width = 0.12) +
  geom_jitter(width = 0.05) +
  labs(
    title = "Cycle Time Distribution by Production Line",
    x = "Production Line",
    y = "Cycle Time (minutes)"
  ) +
  theme_minimal()
```



QUESTION 3 – Proportion Visualization and Choice of Chart

Question

A university analyzes preferred learning mode of 240 students across three departments.

Department Online (%) Blended (%) Face-to-Face (%)

CSE	52	28	20
-----	----	----	----

ECE	45	35	20
IT	30	40	30

Tasks

1. Create a side-by-side bar chart and a 100% stacked bar chart.
2. Explain which chart better supports comparison of individual proportions and why.
3. Explain why separate pie charts may be less effective for comparing departments.
4. Identify one design choice that could make small proportional differences appear larger.

R Code

```
library(ggplot2)
```

```
learning <- data.frame(
  Department = c("CSE", "ECE", "IT"),
  Online = c(52, 45, 30),
  Blended = c(28, 35, 40),
  Face_to_Face = c(20, 20, 30)
)
```

```
learning_long <- data.frame(
  Department = rep(c("CSE", "ECE", "IT"), each = 3),
  Mode = rep(c("Online", "Blended", "Face-to-Face"), 3),
  Percentage = c(
    52, 28, 20,
    45, 35, 20,
    30, 40, 30
  )
)
```

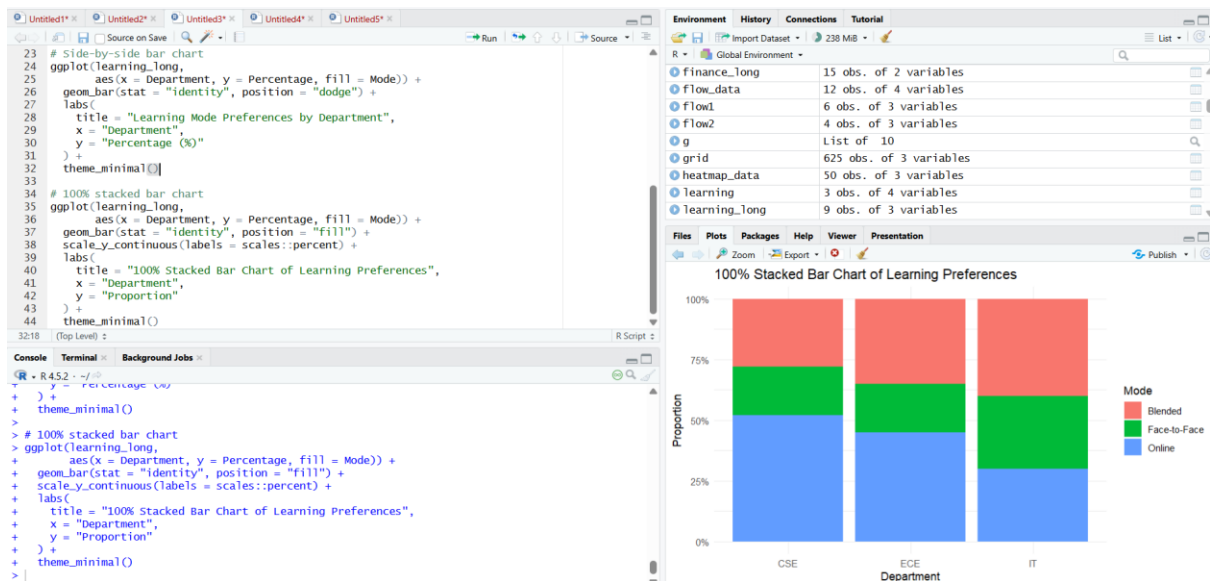
)

Side-by-side bar chart

```
ggplot(learning_long,  
       aes(x = Department, y = Percentage, fill = Mode)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(  
    title = "Learning Mode Preferences",  
    x = "Department",  
    y = "Percentage (%)"  
  ) +  
  theme_minimal()
```

100% stacked bar chart

```
ggplot(learning_long,  
       aes(x = Department, y = Percentage, fill = Mode)) +  
  geom_bar(stat = "identity", position = "fill") +  
  labs(  
    title = "100% Stacked Learning Mode Preferences",  
    x = "Department",  
    y = "Proportion"  
  ) +  
  theme_minimal()
```



QUESTION 4 – Nested Proportions using Treemap and Sunburst

Question

An e-commerce company analyzes annual revenue across product categories and subcategories.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks

1. Construct a treemap showing category and subcategory revenue.
2. Construct a sunburst chart representing the same hierarchy.

3. Determine the largest category and calculate its percentage of total revenue.
4. Compare treemap and sunburst for identifying nested proportions.
5. State one situation in which area encoding could mislead interpretation.

R Code

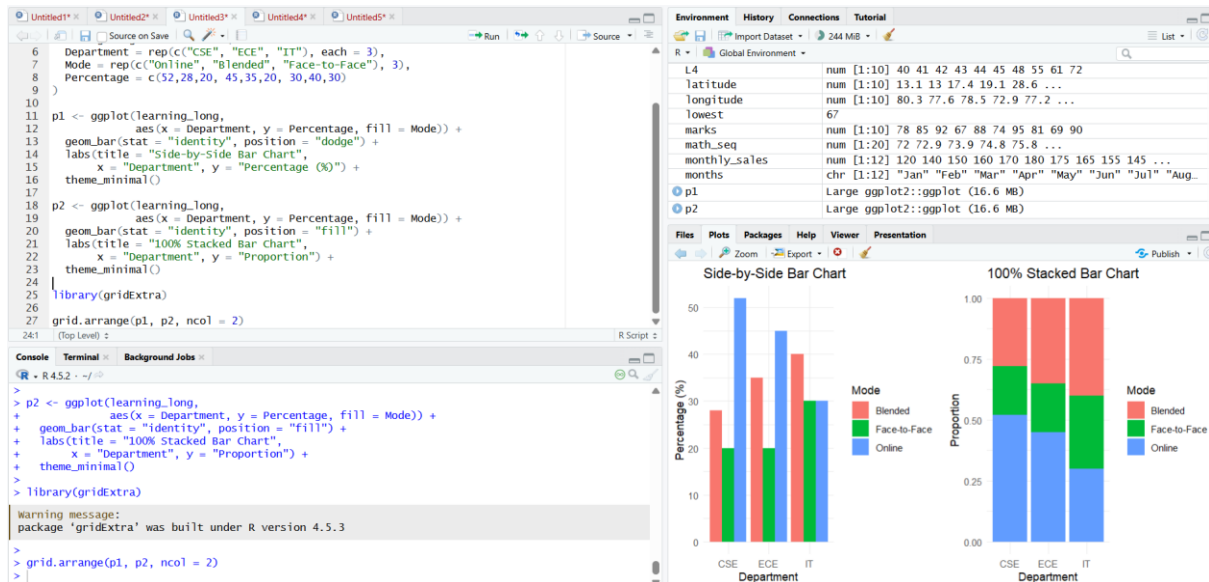
Q3: Side-by-side Bar Chart and 100% Stacked Bar Chart

```
library(ggplot2)
```

```
learning_long <- data.frame(  
  Department = rep(c("CSE", "ECE", "IT"), each = 3),  
  Mode = rep(c("Online", "Blended", "Face-to-Face"), 3),  
  Percentage = c(52,28,20, 45,35,20, 30,40,30)  
)  
  
p1 <- ggplot(learning_long,  
  aes(x = Department, y = Percentage, fill = Mode)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(title = "Side-by-Side Bar Chart",  
    x = "Department", y = "Percentage (%)") +  
  theme_minimal()  
  
p2 <- ggplot(learning_long,  
  aes(x = Department, y = Percentage, fill = Mode)) +  
  geom_bar(stat = "identity", position = "fill") +  
  labs(title = "100% Stacked Bar Chart",  
    x = "Department", y = "Proportion") +  
  theme_minimal()
```

```
library(gridExtra)
```

```
grid.arrange(p1, p2, ncol = 2)
```



QUESTION 4 – Nested Proportions using Treemap and Sunburst

Question

An e-commerce company analyzes annual revenue across product categories and subcategories.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6

Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks

1. Construct a treemap showing category and subcategory revenue.
2. Construct a sunburst chart representing the same hierarchy.
3. Determine the largest category and calculate its percentage of total revenue.
4. Compare treemap and sunburst for identifying nested proportions.
5. State one situation in which area encoding could mislead interpretation.

R Code

Q4: Treemap and Sunburst

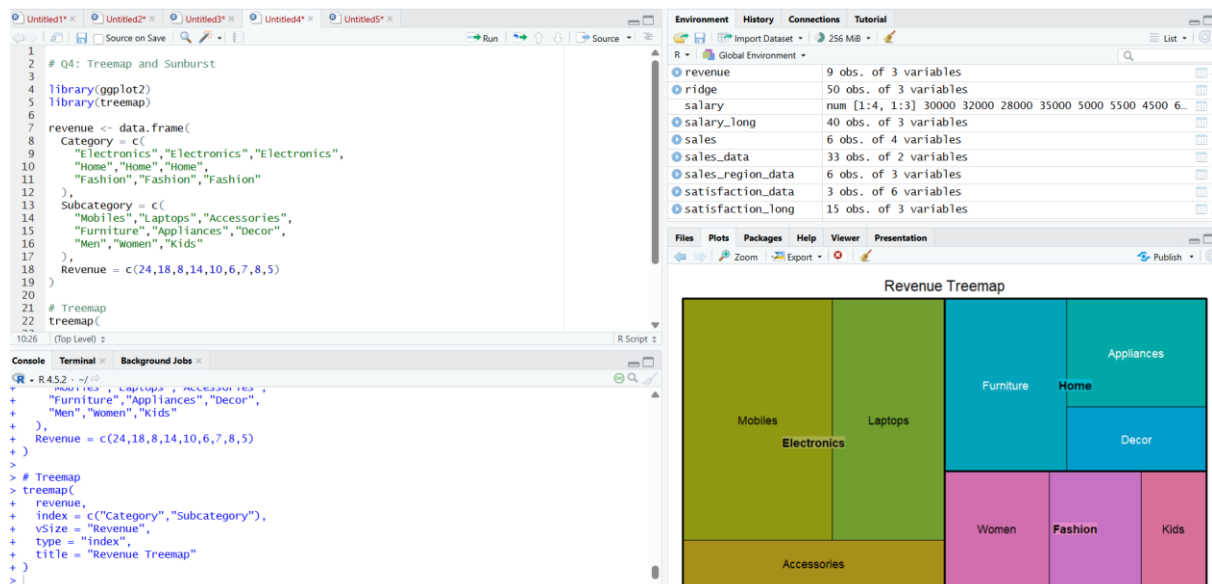
```
library(ggplot2)
```

```
library(treemap)
```

```
revenue <- data.frame(
  Category = c(
    "Electronics", "Electronics", "Electronics",
    "Home", "Home", "Home",
    "Fashion", "Fashion", "Fashion"
  ),
  Subcategory = c(
    "Mobiles", "Laptops", "Accessories",
    "Furniture", "Appliances", "Decor",
    "Men", "Women", "Kids"
  ),
  Revenue = c(24, 18, 8, 14, 10, 6, 7, 8, 5)
)
```

```
# Treemap
```

```
treemap(  
  revenue,  
  index = c("Category","Subcategory"),  
  vSize = "Revenue",  
  type = "index",  
  title = "Revenue Treemap"  
)
```



QUESTION 5 – Flow Proportions using Sankey Diagram / Parallel Sets

Question

A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone.

Origin	Mode	Destination	Commuters
--------	------	-------------	-----------

Residential	Bus	Central	35
-------------	-----	---------	----

Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks

1. Construct a Sankey diagram showing Origin → Mode → Destination.
2. Construct a parallel-sets representation of the same categorical flows.
3. Identify the three largest flows and calculate their percentages of all commuters.
4. Compare Sankey and parallel sets for flow magnitude and category relationships.
5. Explain one way poor link widths or category ordering could mislead the viewer.

R Code

Q5: Sankey Diagram

```
install.packages("networkD3") # Run only once
```

```
library(networkD3)
```

```
flow_data <- data.frame(
```

```
  Origin = c(
```

```
    "Residential", "Residential", "Residential", "Residential",
```

```
    "Suburban", "Suburban", "Suburban", "Suburban",
```

```
"Rural","Rural","Rural","Rural"  
)
```

```
Mode = c(  
  "Bus","Bus","Train","Train",  
  "Bus","Bus","Train","Train",  
  "Bus","Bus","Train","Train"  
)
```

```
Destination = c(  
  "Central","Industrial","Central","Industrial",  
  "Central","Industrial","Central","Industrial",  
  "Central","Industrial","Central","Industrial"  
)
```

```
Commuters = c(  
  35,15,25,10,  
  20,10,30,15,  
  12,8,10,10  
)  
)
```

```
# Create unique nodes  
nodes <- data.frame(  
  name = unique(c(  
    flow_data$Origin,  
    flow_data$Mode,  
    flow_data$Destination  
  ))  
)
```

```
)
```

```
# Origin -> Mode
```

```
flow1 <- aggregate(  
  Commuters ~ Origin + Mode,  
  flow_data,  
  sum
```

```
)
```

```
# Mode -> Destination
```

```
flow2 <- aggregate(  
  Commuters ~ Mode + Destination,  
  flow_data,  
  sum
```

```
)
```

```
links1 <- data.frame(  
  source = match(flow1$Origin, nodes$name) - 1,  
  target = match(flow1$Mode, nodes$name) - 1,  
  value = flow1$Commuters
```

```
)
```

```
links2 <- data.frame(  
  source = match(flow2$Mode, nodes$name) - 1,  
  target = match(flow2$Destination, nodes$name) - 1,  
  value = flow2$Commuters
```

```
)
```

```
links <- rbind(links1, links2)
```

```

sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 30
)

```

